

Overtopping and Local Breach of an Earth Dam under Floods Exceeding the Design Standard

A Technical Review and Industry Lessons from the Liulan Reservoir Event in Hengzhou, Guangxi, China

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Note on sources and scope: This article is a preliminary technical review based on public reports, official updates, and basic principles of hydraulic and water resources engineering available during July 6–8, 2026. It does not substitute for a governmental incident investigation, dam-safety assessment, or formal hydrologic review. Data related to water levels, storage capacity, breach width, flood volume, project condition, and reservoir operation process should be understood as preliminary estimates or reasonable inferences based on public information unless they have been confirmed by an official investigation.

Abstract

On July 6, 2026, Liulan Reservoir in Hengzhou, Nanning, Guangxi, experienced overtopping and breach openings in its dam during a historically rare extreme rainfall event, creating a severe flood emergency. From the perspective of hydraulic and water resources engineering, this article provides a preliminary review of the engineering background, possible hazard mechanisms, reservoir-release constraints, overtopping failure mechanism of the earth dam, secondary sediment and water-quality impacts, post-disaster recovery needs, and directions for improving forecasting and warning systems.

Based on currently available public information, the direct trigger of the event was an above-design-standard inflow flood caused by typhoon rainfall. The vulnerability of an aging homogeneous earth dam to overtopping flow, together with the imbalance between spillway/release capacity and the incoming flood peak, formed an important engineering context for the rapid escalation of the emergency. High downstream exposure of population and infrastructure was also a major factor contributing to the severity of the disaster consequences. This article emphasizes that such events should not be attributed to a single cause before the completion of an official investigation. Instead, hydrologic reconstruction, operation review, dam-safety assessment, and basin-scale risk evaluation should be used to promote a transition in aging-reservoir management from design-standard-based engineering protection to risk-based system management.

1. Project Overview and Functional Role: Basic Profile of Liulan Reservoir

Liulan Reservoir is located in Liulan Village, Xiaoyi Town, Hengzhou City, Nanning, Guangxi, at approximately 22°56'N and 109°12'E. The reservoir impounds runoff from Zhenlong River, a first-order tributary of the Yujiang River; the upstream reach is also referred to as Yunbiao River. The drainage area above the dam site is approximately 195 km². Publicly reported total storage capacity is approximately 93.19 million m³, although another public source gives approximately 95.52 million m³. The dam height is approximately 42 m. It is a medium-sized reservoir serving flood control, irrigation, water supply, and other multipurpose functions.

From an engineering-structural perspective, public information indicates that the main dam of Liulan Reservoir is a homogeneous earth dam, meaning that the dam body was formed by placing and compacting cohesive soil materials in layers. Construction reportedly began in 1958 and was completed in 1960, so the project has been in operation for more than 60 years. Homogeneous earth dams are relatively low-cost and comparatively simple to construct, but they generally are not designed to withstand sustained overtopping flow. Once floodwater flows over the crest, concentrated flow on the downstream slope may rapidly erode and remove dam materials, leading to breach initiation and enlargement.

Public information also mentions that the reservoir historically experienced safety issues such as upstream-slope sliding, locally loose soil, and seepage, and that dam-safety reinforcement works had been undertaken. This article treats such information only as engineering background. The specific condition of historical deficiencies, the scope and quality of reinforcement, and their relationship, if any, to the present event should be determined by formal engineering assessment and official investigation.

In terms of beneficial-use functions, Liulan Reservoir supports agricultural irrigation in Xiaoyi, Yunbiao, Maling, Hengzhou, and other townships. The reported benefited farmland area is about 157,300 mu, and the reservoir is also related to local water supply, irrigation-district modernization, ecological uses, and fishery management. A modernization project for the irrigation district reportedly began in 2023, with a total investment of about RMB 383 million. Local media reported in April 2026 that approximately RMB 290 million had been completed and that civil-works completion had reached about 95%. Improved irrigation water-use efficiency, water-source reliability, and source-water protection are therefore closely connected to reservoir operation. This shows that Liulan Reservoir is not a single-purpose flood-control project but an important node in the regional agricultural, water-resources, and livelihood system.

During recent severe droughts, Liulan Reservoir reportedly released water to support spring irrigation. This fact indicates that post-disaster decisions on whether and how to repair the reservoir cannot be limited to closing the breach. They should also consider regional water supply and demand, irrigation-district investments, alternative water sources, engineering safety, and acceptable risk from multiple perspectives.

2. Hazard Mechanisms: Above-Design Floods and Overtopping Vulnerability of an Earth Dam

2.1 Extreme rainfall and inflow-magnitude assessment

The direct trigger of this emergency was extreme rainfall associated with Typhoon Maysak, the tenth typhoon of the year. Public reports indicate that from 08:00 on July 5 to 08:00 on July 6, 2026, Nanning and surrounding areas experienced rainstorms to extreme rainstorms. The 24-hour rainfall reached 565.0 mm at Shitang Town, 445.0 mm at Taoxu Town, and 411.5 mm at Xiaoyi Town in Hengzhou. At Bafeng Village, Luxu Town, Binyang County, the 24-hour rainfall reached 713.3 mm; its 3-hour, 6-hour, 12-hour, and 24-hour rainfall totals all broke historical records for Nanning, and the 24-hour total ranked second in the historical record for Guangxi. The Hengzhou–Binyang–Guigang area was near the center of the intense rainfall, and regional river levels rose rapidly.

Using the approximately 195 km² drainage area of Liulan Reservoir for an order-of-magnitude estimate, it should be noted that the Shitang station may not be within the reservoir catchment or may be located only near its edge; the measured rainfall at Xiaoyi Town, closer to the core of the catchment, was 411.5 mm. If 411.5–565 mm is used as a representative rainfall range, then the 24-hour total rainfall volume over the catchment would be approximately 80 to 110 million m³. Under conditions of near-saturated antecedent soil, extremely high rainfall intensity, and high runoff efficiency, assuming a runoff coefficient of 0.75–0.85 could yield an inflow volume of approximately 60 to 94 million m³. This magnitude approaches, and may even reach, the total storage capacity of the reservoir.

It must be emphasized that the above calculation is only a rough order-of-magnitude assessment based on point rainfall and a simplified runoff coefficient. It cannot substitute for a formal hydrologic reconstruction. Subsequent review should incorporate basin-average rainfall, radar rainfall, antecedent soil moisture, measured stage-storage relationships, inflow hydrographs, outflow rating curves, and gate-operation records.

2.2 Water-level rise, release capacity, and operation constraints

Public reports indicate that before the incident the water level of Liulan Reservoir had exceeded the design flood level. According to media summaries, the reservoir station had conducted two flood-regulation releases before the event and had all personnel on standby on the afternoon of July 5. Around 02:30 on July 6, the reservoir reportedly exceeded the flood-control limit level and issued multiple release warnings. At 06:00 the gates were fully opened, and at around 10:20 the dam was overtopped and breach openings appeared. Another report stated that the reservoir

water level reached 111.20 m at around 02:30 on July 6, exceeding the design flood level by 0.91 m. That value is not an official record and should be verified against gate-operation records and automatic water-level data. Despite full-gate release, the water level continued to rise and ultimately overtopping occurred. This indicates that, at least during this extreme event, there was a significant imbalance between the inflow-flood magnitude and the project's release capacity.

Reservoir operation should not be simplified as “opening gates earlier would necessarily have prevented overtopping.” The maximum discharge capacity of gates, spillways, and outlet works is jointly constrained by opening size, crest elevation, gate condition, hydraulic head, and downstream conveyance capacity. If the inflow hydrograph far exceeds the project's release capacity, earlier and larger releases can reduce or delay water-level rise but may not fully prevent overtopping. According to water-conservancy gazetteer-type materials circulated online, the normal spillway of Liulan Reservoir may be a shoulder-side-channel practical weir with a crest elevation of 108.10 m and a maximum discharge of about 2,240 m³/s. These parameters have not been officially confirmed, and “design maximum discharge” is not equivalent to the actual usable discharge capacity during this flood event. A later review should verify the actual discharge-capacity curve using design and safety-assessment documents, measured reservoir stage, downstream water levels, gate condition, and any inlet or outlet blockage or obstacles.

At the same time, the possibility of operational optimization cannot be ruled out in the absence of complete operation records. Key questions for review include whether the pre-event reservoir level was relatively high, whether pre-releases had been conducted based on typhoon forecasts, whether maximum discharge capacity was fully used, whether downstream channels and Yujiang River water levels constrained pre-release, and whether downstream warnings and evacuation were coordinated with reservoir operation.

2.3 Overtopping failure mechanism of a homogeneous earth dam

From the perspective of earth-dam engineering, the development of the Liulan Reservoir emergency is consistent with the general pattern of overtopping failure of earth dams. The downstream slope of a homogeneous earth dam is typically protected by grass cover, dry masonry, or other slope-protection measures against rainfall erosion and small surface runoff. Such measures are not designed to withstand sustained, high-discharge overtopping flow.

A possible failure process includes four stages. First, after the reservoir water level exceeds or approaches the dam-crest control elevation, water flows over the crest and down the downstream slope. Second, high-velocity flow damages the slope protection and exposes the dam-fill material. Third, concentrated erosion forms rills and gullies on the slope; these gullies headcut upstream and deepen continuously. Fourth, when a gully cuts through a critical section of the dam body, the remaining dam mass cannot resist the upstream water pressure, and local sliding and breach enlargement follow.

According to public images and reports, at around 10:20 on July 6, two breach openings appeared at the eastern dam head and on the left-side dam section near the power station, with an initial combined length reported at about 50 m. By the evening of the same day, media cited reservoir staff as saying that the breach had expanded to about 100 m wide and that both sides were still collapsing. According to a July 7 report by Guangxi Daily, the overtopping breach at Liulan Reservoir had become stable and was no longer expanding. The precise location, width, bottom elevation, development duration, and stage-by-stage enlargement rate of the breach—as well as whether it was related to dam abutment connections, through-dam structures, a suspected emergency-spillway facility, or local fill quality—should be confirmed by field survey, UAV imagery, dam cross sections, material testing, and operation records. Special attention is needed because media expressions such as “eastern dam head” and “left-side dam section near the power station” involve two different coordinate references: cardinal directions and left/right bank (or observer perspective). By the hydraulic convention of facing downstream, the left bank at this dam site roughly corresponds to the east bank; however, if an eyewitness was looking from the upstream reservoir side, “left side” would be opposite to the hydraulic left bank. Existing text alone cannot close the positional relationship between the two breach openings. A formal review should locate the two breaches using the dam-axis stationing system from design drawings to avoid ambiguity caused by mixing east/west with left/right terminology.

Another issue that should be carefully verified in the formal investigation is the engineering nature and actual operating condition of the open bypass channel outside the east-bank dam shoulder. Pre-breach satellite imagery

shows that a normal spillway can be identified near the right-bank dam shoulder of Liulan Reservoir. At the same time, there is another distinct open bypass channel outside the east-bank dam shoulder, with artificial shaping and features suggestive of an auxiliary release channel or emergency spillway channel (see figure). Because satellite imagery does not provide control elevation, crest structure, design discharge, or activation conditions, this article does not determine the engineering function of that facility. Online sources have also mentioned that Liulan Reservoir may have a fuse-plug-type emergency spillway facility, but its location, elevation, activation criteria, current condition, and relationship to the present breach have not been confirmed by authoritative sources. Subsequent investigation should determine whether this east-bank channel is an auxiliary spillway, emergency spillway, or another engineering feature; whether its design function, actual discharge capacity, and current condition meet design requirements; whether the highest water level in this event reached its activation condition; whether the downstream floodway has been occupied or obstructed; and whether it has any direct relationship with the breach openings. This article does not use this issue to characterize the event; it identifies it only as an important issue for subsequent engineering review.



Figure: Pre-event satellite imagery of Liulan Reservoir engineering layout (Google Maps; imagery source: Imagery ©2026 CNES / Airbus, Maxar Technologies)

Therefore, this article prefers to describe the event as “overtopping and localized dam breach openings” or “local breach following overtopping of an earth dam.” Before official investigation conclusions are released, directly characterizing it as a full dam failure or a single engineering-quality problem would be overly simplistic.

2.4 Sediment transport and secondary water-quality impacts

Overtopping and dam-breach openings not only release floodwater but may also transport large quantities of sediment, organic matter, nutrients, and other pollutants. Images from the site show yellow-brown flow, indicating that the floodwater may have had a high suspended-sediment concentration. Possible sources include watershed surface erosion, resuspension of reservoir sediments, and erosion of dam-fill material.

Under intense rainfall, sloping farmland, disturbed forest areas, road slopes, exposed ground surfaces, and channel banks within the watershed may all become sediment sources. If the reservoir water level drops rapidly after breach formation, reservoir shoals and loose bottom sediments may also be re-eroded and transported downstream. The dam material itself, eroded during overtopping, directly adds to the sediment concentration in the released flow.

Such sediment-laden floodwater may deposit in downstream channels, farmland, roads, and low-lying village or town areas, affecting flood conveyance sections, cultivated soil layers, intake operation, water-quality protection,

and ecosystem recovery. Therefore, post-disaster dredging, sediment handling, water-quality monitoring, environmental sanitation, and necessary public-health disinfection should be incorporated into the recovery plan together with dam repair.

3. Basin Relationships and Related-Project Clarifications

3.1 Flood superposition with Yunbiao Reservoir and simultaneous multi-reservoir emergencies

Yunbiao Reservoir is located about 5 km west of Liulan Reservoir. Official updates indicate that on the morning of July 6, both reservoirs experienced overtopping and breach openings, and their downstream channels merge near Naguang Village before entering the Yujiang River. On the same day, Sanchakou Reservoir and Chayuan Reservoir in Hengzhou also experienced emergencies, and Liuwang Reservoir in Binyang County experienced overtopping of the embankment.

This means that inundation in some downstream towns and villages was the result of superposed flood sources from two or more reservoirs. For example, a village official in Nankang Village, Yunbiao Town, told the media that approximately 6,390 people in the village were affected by floodwater associated with overtopping and breach openings at the two reservoirs, and nearly 1,700 people were at one point surrounded by floodwater. Therefore, in disaster attribution, inundation reconstruction, and loss assessment, downstream flooding should not be simply attributed to a single breach source from Liulan Reservoir. The fact that multiple reservoirs in the same region experienced emergencies on the same day is itself important evidence for the “systemic risk” argument presented in Section 9.

3.2 Relationship with the Datengxia Water Control Project

Liulan Reservoir is located upstream on Zhenlong River, a tributary of the Yujiang River. Floodwater from the reservoir enters Zhenlong River, then the Yujiang River, and eventually the Xijiang River system. The Datengxia Water Control Project is located on the Qianjiang mainstem in Guiping City. The two projects are hydrologically connected within the same basin, but they are not in a close upstream–downstream relationship.

From the perspective of hydraulic propagation, backwater from the Datengxia reservoir area could not propagate upstream to the river reach below Liulan Reservoir and cause this event. The river distance is long, and the Liulan emergency occurred at the upstream tributary reservoir itself. Public information indicates that Datengxia undertook regulation and flood-peak attenuation during this regional flood process. The specific flood volume retained, peak-reduction effect, and operational role should be determined by water-authority operation reports.

3.3 Relationship with the Pinglu Canal

The starting point of the Pinglu Canal is located in the Xijin reservoir area of the Yujiang mainstem within Hengzhou. It is in the same regional Yujiang water-system context as Liulan Reservoir, but the two are not on the same direct flood-conveyance path. Floodwater from Liulan Reservoir enters the Yujiang via Zhenlong River and then generally propagates downstream along the Xijiang River system; the Pinglu Canal is an artificial channel connecting the Yujiang southward to the Beibu Gulf.

The actual hydraulic role of the Pinglu Canal during this flood cannot be determined merely from geographic proximity. Whether it affected local water levels, flow distribution, or flood-conveyance conditions should be evaluated using canal-control operation records, Yujiang stage-discharge hydrographs, and hydrodynamic modeling. This article only notes the regional water-system connection and does not treat the Pinglu Canal as a cause of the Liulan Reservoir emergency.

3.4 Reconsideration of earthquake influence

After the May 2026 earthquake in Liuzhou, Guangxi, some members of the public raised concerns about whether seismic activity may have affected regional reservoir safety. Such concern is reasonable. For the Liulan Reservoir event, however, public information has not indicated that the earthquake was a direct or major cause of the dam-

breach openings. The direct trigger of this emergency remains extreme rainfall and the resulting above-design-standard flood and overtopping process.

Nevertheless, the combined risk of seismic and flood conditions deserves attention. For aging earth dams, earthquakes may create microcracks, local loosening, or changes in seepage conditions. Under extremely high reservoir levels, increased seepage pressure may accelerate piping or local instability along such weak zones. Although these effects may not have been the primary cause in this event, they remind the industry that old reservoirs, deficient reservoirs, and projects in complex geological areas require special attention during pre-flood inspections, post-earthquake inspections, and risk reassessment before intense rainfall.

4. Source-Water Protection, Water Quality, and Watershed Management

In addition to irrigation and flood-control functions, Liulan Reservoir is also an important regional water source. Some public materials have mentioned source-water management issues in the reservoir area and upstream watershed, including elevated total nitrogen, fertilizer input to the reservoir, soil erosion, and fast-growing eucalyptus plantations. These issues are not part of the same causal chain as the present dam-breach openings, but they are related to runoff generation, sediment production, water-quality risk, and post-disaster recovery under extreme rainfall, and they deserve continued attention in long-term watershed management.

During extreme rainfall, watershed nonpoint-source pollution, resuspension of reservoir sediments, and wash-off of pollutants from farmland, villages, and towns may all increase downstream water-quality risk. Post-disaster recovery should therefore not focus only on structural dam repair. It should also include inspection of source-water protection zones, monitoring of sediments and pollutants, reservoir-bank stability checks, restoration of ecological buffer zones along inflowing rivers, and upstream hillslope soil-and-water conservation measures.

5. Post-Disaster Recovery: A System Project Beyond Breach Repair

5.1 Repair of hydraulic works and infrastructure

Post-disaster recovery should first include dam-safety assessment and breach treatment. For a breach on the order of about 100 m, topographic survey, geotechnical investigation, residual-dam stability analysis, seepage analysis, and emergency diversion planning should be conducted before determining temporary closure, permanent repair, dam raising, or construction of new flood-release facilities.

At the same time, rapid reservoir drawdown may induce reservoir-bank instability. Reservoir-bank failures, slope cracks, road damage, and risks to structures around the reservoir should be inspected. Downstream channels may experience severe sediment deposition, bank erosion, local scour at bridge piers, culvert blockage, roadbed undermining, and damage to weak levee sections; all of these require systematic safety assessment.

Power supply, water supply, communications, roads, and bridges form the basis of post-disaster life recovery. Transformers, pumping stations, cables, communication facilities, and water-supply pipelines that were inundated by floodwater should not simply be returned to service. They should be safety-tested, disinfected, flushed, and put through trial operation before gradual restoration.

5.2 Dredging, ecological recovery, and public-health restoration

After sediment-laden floodwater deposits in downstream farmland, towns and villages, roads, and channels, it may affect cultivated soil layers, drainage capacity, and public health. Dredging should distinguish among flood-conveyance dredging in channels, road and village/town cleanup, farmland topsoil restoration, and disposal of potentially contaminated sediments. Not all sediment should be treated simply as ordinary silt.

If reservoir-bottom sediment contains relatively high organic matter, nitrogen, phosphorus, or other pollutants, post-disaster water-quality and sediment monitoring should also be conducted. Drinking-water sources, temporary intakes, wells, and inundated water-supply facilities should be tested for microorganisms, turbidity, organic matter, and necessary chemical indicators to ensure public-health safety.

5.3 Assistance, compensation, and rebuilding public confidence

According to the Nanning flood-control and disaster-relief press conference on July 7, as of that morning, Hengzhou had 4 deaths, 8 missing persons, 84,700 affected people, and 53,808 people urgently evacuated and relocated; Binyang County had 8,606 affected people, and all persons requiring evacuation had been evacuated and relocated. The evening Guangxi regional emergency-response press conference on July 7 further reported that 375,000 people across the region had been affected, 6 had died, 11 were missing, and 130,000 had been urgently evacuated and relocated. These figures are dynamic statistics, and final values should be based on official verification.

In the short term, disaster assistance should focus on shelter, drinking water, hot meals, medical care, disinfection, transportation restoration, and delivery of relief supplies. In the medium term, it should focus on housing, farmland, roads, hydraulic facilities, and public-service systems. In the long term, the focus should be risk communication, disclosure of engineering safety assessments, repair-scheme evaluation, and clarification of responsibility boundaries.

Rebuilding public confidence cannot rely only on explaining that “this was caused by extreme weather.” More important is allowing affected communities to see concrete actions in dredging, emergency repair, compensation, relocation support, risk assessment, and long-term governance. The role of professional media and technical experts is to help the public understand complex causality and avoid reducing the event to either a single engineering responsibility or a single natural disaster.

5.4 Public water-contact safety warning

After the flood from the breach recedes, the downstream river channel does not immediately become safe. High-velocity flow may form deep pits, submerged grooves, and abrupt drops in the channel bed. Large amounts of dam material, trees, building debris, and household waste may be deposited underwater. Riverbank toes may have been scoured and remain unstable. Bridges, culverts, and levees may also contain hidden damage.

Therefore, local emergency-management and water-resources authorities should issue clear channel-safety warnings, place visible warning signs at dangerous reaches, and repeatedly notify residents through village broadcasts, new media, schools, and community channels. Before systematic safety assessment, debris removal, and water-quality testing are completed, swimming, wading, boating, fishing, and activities near damaged bridges or culverts should be strictly prohibited. Safety education for children and adolescents should be especially strengthened.

6. Forecasting and Warning System: The “Four Pre-” Direction and Improvement Needs

This event indicates that traditional reservoir-operation and risk-communication systems still have shortcomings when facing extreme events that exceed design standards. Many aging reservoirs have limited monitoring intelligence, and data-transmission stability may decline during extreme weather. Samples of above-design-standard floods are rare, hydrologic forecasting is difficult, and whether warnings can reach threatened populations rapidly and accurately directly affects evacuation outcomes.

In recent years, the water sector in China has emphasized the “four pre-” system: forecasting, warning, rehearsal/simulation, and emergency plans. For aging reservoirs and small-to-medium watersheds, the key is not merely installing more sensors, but building an integrated chain from rainfall forecasting, watershed runoff generation and routing, reservoir inflow, reservoir operation, downstream inundation, population evacuation, and emergency response.

Technically, radar rainfall, numerical weather prediction, automatic water-level and rainfall stations, watershed hydrologic models, reservoir-operation models, machine learning, and digital-twin platforms can be integrated to improve linked forecasting of extreme rainfall, floods, and reservoir operations. AI and large-model technologies should be treated as supporting tools, with emphasis on data integration, scenario rehearsal, risk identification, and

emergency-information organization, rather than on simple claims of fixed-percentage improvements in forecast accuracy.

Institutionally, clearer graded response procedures should be developed for floods exceeding the design standard: when to pre-release, when to lower the flood-control limit level, when to initiate downstream evacuation, when to enter dam-protection emergency mode, and when to use emergency spillway facilities or temporary flood-diversion measures. Emergency plans should be exercised regularly and linked to evacuation routes for townships, villages, schools, elderly-care institutions, and transport nodes.

7. Repair Necessity and Scheme Comparison Based on Functional Tradeoffs

Liulan Reservoir has multiple functions, including irrigation, water supply, flood control, power generation, ecology, and fisheries. Discussion of whether and how to repair the reservoir should not be simplified as a question of restoring the original condition. A more reasonable approach is to compare schemes based on functional redundancy, feasibility of alternative projects, downstream risk, and future climate pressure.

Function	Main consequences if not repaired	Feasibility of alternatives	Repair urgency
Irrigation	Water-source security for about 157,300 mu of farmland would be affected, and the benefits of completed irrigation-district modernization investments would be difficult to realize.	Regional water transfers, ponds, pumping stations, or temporary water sources could partially substitute, but complete short-term substitution would be difficult.	Extremely high
Water supply	Water-supply security and emergency water-source capacity for surrounding townships would decline.	Switching to or constructing alternative water sources is possible, but supporting pipelines, pumping stations, and water-quality safeguards would be required.	High
Flood control	Upstream regulation and storage capacity would be weakened; however, this event also exposed the inadequacy of the original flood-control capacity.	Flood-control standards, emergency spillway facilities, downstream levees, and warning/evacuation systems would need to be reassessed.	Requires reassessment
Power generation	Small hydropower revenue would be lost.	The regional energy system can substitute; this function has relatively low priority.	Low
Ecology and fisheries	The aquatic ecosystem and management practices such as “using fish to improve water quality” would be interrupted.	They can be gradually rebuilt as the reservoir and water quality recover.	Moderate

The above framework indicates that Liulan Reservoir has a very high need for repair, mainly because its irrigation and water-supply functions are difficult to fully replace in the short term. However, repair should not simply restore the original condition. Instead, the event should be treated as a window of opportunity for functional reallocation, standard improvement, and risk transfer.

At least three options should be compared. Scheme A would repair the reservoir at the original site while improving flood-control and overtopping-resistance capacity. Scheme B would repair the reservoir but adjust its functional role, lower high-risk beneficial-use water levels, or operate it at a downgraded level. Scheme C would partially

abandon the existing reservoir and build alternative water-supply or alternative storage/regulation works. The final choice should comprehensively consider engineering safety, economics, downstream population risk, irrigation and water-supply security, ecological impacts, and long-term operation and maintenance capacity.

If the original-site repair option is selected and the reservoir continues to serve important water-supply and irrigation functions, key issues should include whether to raise the dam crest and increase wave freeboard, whether to expand or construct spillways, whether to review the reliability, control elevations, activation conditions, and downstream land-use controls of existing spillway facilities and suspected emergency spillway facilities, whether emergency spillway facilities and erosion-resistant linings should be rebuilt, expanded, or added, whether cutoff walls, drainage prisms, and slope reinforcement should be adopted, whether automatic rainfall and water-level monitoring and remote gate-control systems should be upgraded, and whether downstream flood-risk zones and emergency evacuation routes should be redelineated.

8. Subsequent Research and Work Recommendations

The research and technical work listed in this section directly supports the repair-necessity assessment and scheme comparison in Section 7. Whether the final decision is original-site repair, downgraded operation, or an alternative scheme, the following hydrodynamic reconstruction, risk-chain analysis, risk ranking, and multi-objective comparison studies are needed as a basis for decision-making.

8.1 Quantitative simulation of breach-flood propagation

A one-dimensional/two-dimensional hydrodynamic model should be established for the reach from Zhenlong River to the Yujiang River, using HEC-RAS, MIKE, or a similar model. Because Yunbiao Reservoir also experienced overtopping and breach openings on the same day, and because the downstream channels of the two reservoirs merge, the model should use dual-source inflow boundaries for Liulan and Yunbiao, and multi-source boundaries if necessary. Otherwise, inundation reconstruction for Xiaoyi, Yunbiao, Maling, and nearby downstream areas may contain systematic bias. The model should use measured or verified storage curves, breach width, breach invert elevation, breach development time, and water-level process to back-calculate breach-flood hydrographs and quantitatively evaluate inundation extent, depth, velocity, and arrival time in downstream townships.

If conditions permit, the model should further couple a sediment-transport module to assess deposition of sediment-laden floodwater in channels, farmland, and villages/towns. This would provide quantitative support for channel dredging, bridge and culvert inspections, and farmland recovery.

8.2 Severe-consequence risk chain and nonstationary frequency analysis

A complete risk-chain analysis framework should be established, for example: $P(\text{severe consequences}) = P(\text{flood exceeds design standard}) \times P(\text{overtopping} \mid \text{above-design flood}) \times P(\text{breach formation} \mid \text{overtopping}) \times P(\text{exposure of people and property} \mid \text{flood release})$. Such a framework helps place “extreme rainfall,” “engineering vulnerability,” “operation capability,” and “downstream exposure” within a unified risk-management system.

Under climate change, traditional “100-year return period” frequency estimates may no longer satisfy nonstationary extreme-rainfall conditions. Regional rainfall records, changes in typhoon tracks, extreme short-duration rainfall intensity, and future climate scenarios should be used to conduct nonstationary frequency analysis and adaptive assessment of flood-control standards.

8.3 Risk ranking of aging earth dams and reinforcement pathways

A regional risk-ranking inventory for aging earth dams should be established using indicators such as dam height, storage capacity, downstream population, project age, dam type, release capacity, historical deficiencies, geological conditions, sensitivity to intense rainfall, and management capacity. For projects with high risk levels, safety assessment, flood-regulation capacity review, spillway-capacity assessment, and downstream flood-risk mapping should be prioritized.

Reinforcement pathways should not be limited to seepage control of the dam body. They should also include improving release capacity, dam-crest wave protection and anti-overtopping measures, erosion-resistant

downstream slope protection, monitoring automation, gate reliability improvement, and emergency access construction. For small projects with low necessity or unacceptably high risk, downgraded operation, retirement, or replacement should also be considered.

8.4 Strategic comparison of repair schemes

The post-disaster repair of Liulan Reservoir should involve systematic scheme comparison. Original-site repair with increased standards would help preserve irrigation and water-supply functions, but it would require substantial investment and renewed assessment of above-design flood risk. Downgraded operation could lower beneficial-use water levels and downstream risk, but it may weaken water supply and irrigation capacity. Construction of alternative water sources or alternative storage/regulation projects would have a long cycle and high coordination difficulty.

Multi-objective decision analysis should use criteria such as life-cycle cost, risk-reduction benefits, functional-security capacity, ecological impacts, social acceptability, and operation and maintenance capacity.

8.5 Ecological and public-health impact assessment

Post-disaster ecological and public-health impact assessment should be conducted. Key issues include the effects of downstream sediment deposition on benthic organisms and fish habitat, sediment-cover thickness on farmland and changes in soil fertility, release risks from nitrogen, phosphorus, organic matter, and possible pollutants in reservoir-bottom sediments, and drinking-water safety and vector-borne disease risks in affected areas.

9. Industry Lessons: From Design Standards to Risk Management

The core lesson of the Liulan Reservoir event is that an aging homogeneous earth dam encountered extreme rainfall exceeding its design standard, and its flood-regulation capacity, resistance to overtopping, and downstream emergency-response system were all tested under limiting conditions. Extreme rainfall was the direct trigger; overtopping vulnerability of the earth dam was a key engineering mechanism; and reservoir operation, anti-overtopping capacity, downstream warning and evacuation, and post-disaster recovery systems jointly determined the event consequences. During the same typhoon rainstorm, several reservoirs, including Yunbiao, Sanchakou, Chayuan, and Liuwang, experienced emergencies on the same day. This further indicates that the event was not an isolated single-reservoir incident but a manifestation of regional water-infrastructure systems under above-design loading. According to the Guangxi regional press conference on the evening of July 7, at that time there were still 341 reservoirs above flood-control limit levels and 56 hydrologic stations on 41 rivers above warning levels. This illustrates the overall stress on the regional water system.

A “100-year” design standard does not mean that a project will necessarily be safe for 100 years. If it is interpreted in simplified terms as a 1% annual exceedance probability, then a project designed for a 100-year event has approximately a $1 - (0.99)^{50} \approx 39.5\%$ probability of experiencing an event exceeding that standard during a 50-year operating life. Therefore, a design standard is the result of engineering-economic and risk tradeoffs, not an absolute safety guarantee.

When a flood exceeding the design standard occurs, reservoir operation objectives shift from normal beneficial use, flood control, and safe release to the extraordinary objectives of dam protection, loss reduction, warning, and population evacuation. For aging earth dams, response pathways for above-design floods must be clarified in advance. These include pre-release rules, emergency spillway measures, emergency materials, emergency transportation, power supply, downstream evacuation thresholds, and public risk communication.

Against the background of climate change and intensified extreme weather, hydraulic-infrastructure safety management needs to shift from “probability-based engineering design” to “risk-based system management.” This means the industry should not only ask whether a project meets existing design standards, but also how it will fail when those standards are exceeded, how floodwater will propagate after failure, how downstream people will evacuate, how infrastructure will recover, and whether there are lower-cost risk-transfer pathways.

For the large number of homogeneous earth dams and small-to-medium reservoirs built across China in the 1950s–1970s, the Liulan Reservoir event provides an important warning. Future safety reinforcement should not remain at the minimum level of “filling gaps.” It should incorporate new evidence on extreme rainfall, downstream socioeconomic exposure, and digital monitoring capacity to reassess safety redundancy, flood-regulation capacity, and emergency resilience.

Sources

The information in this article is mainly from official updates, publicly released meteorological and water-authority information, and domestic public media reports from July 6–8, 2026, including Xinhua News Agency, CCTV, The Paper, Jimu News, Yicai, Red Star News, Southern Wind, Qilu Evening News, and Time Weekly. Engineering parameters related to the spillway, the suspected emergency spillway facility, and discharge capacity are taken from water-conservancy gazetteer-type materials circulated online and remain to be verified. Engineering and disaster data in this article may be revised as formal investigation reports become available.

AI-Assisted Writing Statement

This article was conceived and written by the author, who is responsible for all content. Artificial intelligence tools were used during the writing process to assist with public-information retrieval, fact checking, order-of-magnitude estimate verification, and language revision. All technical judgments and conclusions in the article were reviewed and confirmed by the author, who bears full responsibility for the text.

Appendix 1. Key Data for Liulan Reservoir (Compiled from Public Information)

Item	Data or description
Location	Liulan Village, Xiaoyi Town, Hengzhou City, Nanning, Guangxi
Geographic coordinates	Approximately 22°56'N, 109°12'E; the intake coordinates are separately reported in public materials as 109°16'10"E, 22°56'02"N
Water system	Zhenlong River, a tributary of Yujiang River in the Xijiang River system of the Pearl River Basin
Drainage area	Approximately 195 km ²
Total storage capacity	Public sources report two values, approximately 93.19 million m ³ and approximately 95.52 million m ³ . These may correspond to different statistical or assessment versions and should be confirmed by formal documents.
Dam height	Approximately 42 m
Dam type	Homogeneous earth dam
Construction period	Construction began in 1958 and was completed in 1960
Design standard	Public sources describe it as a 100-year design standard; the specific standard and review results should be confirmed by design and safety-assessment documents.
Main functions	Irrigation, water supply, flood control, and other multipurpose uses
Irrigated area	Approximately 157,300 mu

Water-supply service population	Public sources state approximately 210,000 people; another official wording says the reservoir “benefits more than 170,000 people.” The difference may reflect different statistical categories (water-supply service versus irrigation beneficiaries) and should be verified against local water-supply system data.
Hydropower capacity	Public sources state approximately 1,300 kW

Appendix 2. Public Water-Contact Safety Warning for Downstream Channels

- **Changed riverbed:** High-velocity floodwater may form deep pits, submerged grooves, and abrupt drops that are difficult to see and can easily lead to drowning.
- **Underwater deposits:** Dam material, broken trees, building debris, and household waste may create underwater obstacles and entanglement hazards.
- **Water-quality pollution:** Floodwater may carry bacteria, viruses, pesticide residues, and decaying organic matter. Contact may cause skin disease or gastrointestinal infection.
- **Riverbank collapse:** After floodwater scours the bank toe, riverbanks may still collapse suddenly during the drawdown period.
- **Damaged infrastructure:** Bridges, culverts, levees, and roadbeds may have hidden damage and should not be approached without authorization.

Before systematic safety assessment, debris removal, dredging, and water-quality testing are completed, residents along the river should not enter the channel for swimming, boating, fishing, or playing in the water. Schools, village committees, and communities should strengthen safety education for children.